MILITARY SPECIFICATION

OHMMETERS, INSULATION-RESISTANCE-INDICATING, PORTABLE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 This specification covers portable insulation resistance-indicating ohmmeters of the hand-driven generator type for use in determining insulation resistance of electric circuits and equipment.

1.2 Classification. - The ohmmeters shall be furnished in the following types, as specified (see 6.2):

Type GC - General purpose, 0 to 100 megohm range, 500 volt source, 1/ constant voltage, hand-driven generator type.
Type GM - General purpose, 0 to 1000 megohm range, 500 volt source, 1/ constant voltage, hand-driven generator type.
Type PC - Pocket size, 0 to 100 megohm range, 500 volt source, 1/ hand-driven generator type.
Type PM - Pocket size, 0 to 1000 megohm range, 500 volt source, 1/ hand-driven generator type.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-M-14 - Molding Plastic, and Molded Plastic Parts; Thermosetting.
MIL-P-116 - Preservation, Methods of.
MIL-V-1137 - Varnish, Electrical-Insulating (for Electro-Motive Equipment).
MIL-P-15037 - Plastic-Material, Laminated, Thermosetting, Sheets, Glass-Cloth, Melamine-Resin.
MIL-M-16616 - Manual, Technical, Electronic Equipment.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts: Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

AMERICAN STANDARDS ASSOCIATION
PUBLICATION
C39.1 Requirements For Electrical Indicating Instruments.

(Application for copies should be addressed to American Standards Association, Inc., 70 East Forty-fifth Street, New York 17, N. Y.)

1/ The 500 volt rating is nominal and deviations from this value are acceptable provided the tests of section 4 are met.

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3. REQUIREMENTS

3.1 Preproduction samples.- Preproduction samples of portable insulation resistance indicating ohmmeters, as specified (see 6.2) shall be furnished for preproduction inspection (see 4.3) and shall meet the requirements specified herein. Samples shall be representative of the parts, materials, and processes that will be used by the supplier in production.

3.2 Material.- Material shall be as specified herein. Material not definitely specified shall be of the highest quality and free from defects and imperfections which may affect the serviceability of the finished product. Materials shall be selected on their inherent inability to support fungus growth.

3.2.1 Plastic materials.-

3.2.1.1 Molded thermosetting.- Molded thermosetting plastic shall conform to Specification MIL-M-14.

3.2.1.2 Laminated thermosetting.- Laminated thermosetting plastic material used for identification plates shall conform to Specification MIL-P-15037.

3.2.2 Metal parts.- Metal parts, fittings, and screws not entering into the magnetic circuits shall, where practicable, be made of suitable corrosion resisting material. Magnets and other steel and iron parts entering into the magnetic circuit shall, where practicable, be treated in a satisfactory manner to resist corrosion.

3.2.3 Fungistatic treatment.- Coils having an applied voltage of less than 30 volts and a wire diameter of 0.032 inch, or less, and coils having an applied voltage of 30 volts and higher shall be given three impregnating coats of varnish conforming to type M, grade BB or grade CB, of Specification MIL-V-1137. The coats shall be applied by one of the following methods of impregnation so as to fill all interstices in the coils to preclude the possibility of entrapped air or moisture:

Vacuum-pressure.
Immersion.
Centrifugal.
Pulsating pressure.
Built-up.
Brushing or spraying (for sealer coat only).

3.2.3.1 When required circuit elements such as cable and wire (except molded plastic rubber, and bitumen-coated types), terminal blocks, junction boxes, capacitors, and insulation shall be coated with a moisture and fungus resistant material as specified by the procuring activity (see 6.2).

3.3 Design and construction.-

3.3.1 Balancing.- The moving element shall be balanced by a means which will provide an easy method of rebalancing, such as threaded lock-or split nuts, or a wire helix moving along a cross arm and counter-weight arm. The use of solder, or an adhesive such as shellac, to hold the balance weights or to function as a part of the balance weight will not be acceptable.

3.3.2 Threaded parts.- All threaded parts shall be in accordance with Handbook H28. Whenever practicable, all threads shall be in conformance with the coarse thread series. The fine thread series shall be used only for applications that might show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National form.

3.3.3 Locking of screw-thread assemblies.- When practicable, split-type lock washers or equivalent means shall be provided under all nuts.

3.4 Design and construction of component parts.-

3.4.1 Basic indicating meter.-

3.4.1.1 Types GC and GM.- The ohmmeters shall be of such design that the indication on the scale is independent of the exact voltage applied, and depends only on the resistance under test.

3.4.1.2 Types PC and PM.- The basic indicating ohmmeter design shall either be identical to 3.4.1.1 or designed in accordance with the requirements of American Standards Association Publication C39.1.

3.4.2 Case.- The case shall be of suitable corrosion-resisting metal or of plastic material conforming to Specification MIL-M-14. The case shall be dust-tight and shall enclose all parts, except the crank. The indicating meter mechanism shall be contained within a dust-tight compartment forming a part of the case. The case for type GC or GM ohmmeters shall have a metal or leather handle and shall be provided with a hinged cover. If a crank is provided, it shall not protrude beyond the case by more than 3/4 inch, and may be of the folding type.

3.4.3 Window.- Ohmmeters shall be provided with a window of glass or other transparent material, free from discoloration, scratches, straie, electrostatic effect, or other detrimental defects which may interfere in any way with normal reading of the ohmmeter in daylight. Cement alone is not considered satisfactory for holding the window in place.

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3.4.4 Dial.- The dial shall be made of stiff material firmly secured to its mounting. The dial shall be made of such material so secured as to insure freedom from warping, fading, or other deterioration during the tests specified herein.

3.4.5 Pointer.- The ohmmeter shall have a pointer so formed as to permit accurate readings at the usual distance from which the ohmmeter is viewed and approximate readings up to the distance of legibility of the scale markings. The pointer shall be of rigid construction and shall not become damaged or distorted when subjected to any of the tests specified herein. The distance between the pointer index and the graduated scale shall not exceed 0.10 inch.

3.4.6 Bearings and pivots.- Pivots for moving elements shall be of steel or other suitable material and shall bear upon bearings of equally suitable material.

3.4.7 Terminals.- Binding post terminals shall be constructed so that they cannot turn or twist in their mounting. These terminals shall have non-removable insulated caps with no live parts exposed on cap nuts. Two binding post terminals are required for types GC, PC and PM ohmmeters; three are required for type GM ohmmeters. The third terminal shall be the guard for elimination of leakage errors.

3.4.8 Generators.- Generators shall be smooth running and shall not require starting and running torques which will interfere with practical use. The speed at which it is necessary to turn the crank shall not exceed 170 revolutions per minute. When the armature is not designed for operation in both directions, the crank shall drive the armature shaft by means of a overrunning clutch or other suitable device in order to prevent the armature from being rotated in the wrong direction, and to permit it to run free when the operator ceases to turn the crank. If the armature can be operated in either direction without affecting the instrument deflection, a overrunning clutch will not be required.

3.4.9 Governor.- The generator shall be equipped with a governor or other suitable means for maintaining constant output voltage.

3.4.10 Test leads.- Test leads each approximately 10 feet in length, shall be furnished with each ohmmeter. Each lead shall terminate at one end in an insulated test clip. The other end of each lead shall be provided with a brass spade clip for making positive connection to the ohmmeter terminals. Each spade clip shall be provided with an insulating sleeve over the cord connection joint.

3.4.11 Rectifiers.- Rectifiers shall be of the best commercial quality.

3.5 Interchangeability.- The minimum practicable variety of replaceable parts shall be used throughout the ohmmeter. For example, fixed resistors of the same resistance, tolerance, power rating, ambient temperature rating, grade, physical dimensions and construction, shall be used wherever practicable. However, performance, reliability, long periods of service without maintenance or adjustment, and durability of the equipment shall not be sacrificed to provide the minimum different kinds of replacement parts. Construction of the ohmmeter shall be such that any part, such as a resistor, within the limiting tolerances, characteristics and ratings specified by the applicable specification for the part concerned, may be used to replace the corresponding part in the ohmmeter without modification of the part or the ohmmeter and without causing the ohmmeter to deviate from this specification.

3.6 Types GC and GM.-

3.6.1 Range.-

3.6.1.1 Type GC.- Type GC ohmmeters shall operate in the range of 0 to 100 megohms.

3.6.1.2 Type GM.- Type GM ohmmeters shall operate in the range of 0 to 1000 megohms.

3.6.2 Size.- The maximum overall dimensions shall not exceed 11 inches and the cubical content shall not exceed 650 cubic inches.

3.6.3 Weight.-

3.6.3.1 Type GC.- The weight of the type GC meter shall not exceed 12 pounds.

3.6.3.2 Type GM.- The weight of the type GM meter shall not exceed 13 pounds.

3.6.4 Scale length.- The scale length shall be 2-1/2 inches minimum. Scale length is defined as the distance along the scale arc between the marks corresponding to zero and the specified maximum range of the meter.

3.6.5 Accuracy.- The pointer shall indicate within 1 percent of scale length.

3.6.6 Life.- After the life test specified in 4.6.7, the generator brushes and commutator shall be only slightly worn and there shall be only a slight deposit of carbon or metallic dust in the generator compartment. When in operation, the pointer shall not vary more than 1.0 percent (for

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type GC) and 2.0 percent (for type GM) of scale length from the initial position at any time during the test.

3.6.7 Instruction plate. - Types GC and GM ohmmeters shall be furnished with an instruction plate giving adequate directions for the necessary adjustment and operation of the ohmmeter and directions for the replacement of repair parts, to be securely fastened inside the case cover of the ohmmeter. The plate may be of paper or other suitable material protected by a satisfactory durable, transparent shield or may be of corrosion-resistant metal having the direction stamped, etched, or otherwise indelibly printed thereon. The plate shall also show the terminal voltage and the type, model number, and manufacturer's name of replacement parts that may be used with the ohmmeter.

3.7 Types PC and PM. -

3.7.1 Range. -

3.7.1.1 Type PC. - Type PC ohmmeters shall operate in the range of 0 to 100 megohms.

3.7.1.2 Type PM. - Type PM ohmmeters shall operate in the range of 0 to 1000 megohms.

3.7.2 Size. - The maximum overall dimensions shall not exceed 7-1/2 inches and the cubical contents shall not exceed 225 inches.

3.7.3 Weight. - The weight shall not exceed 3-1/2 pounds.

3.7.4 Scale length. - The scale length shall be 2-11/32 inches minimum. Scale length is defined as the distance along the scale arc between the marks corresponding to zero and the specified maximum range of the meter.

3.7.5 Accuracy. - The pointer shall indicate within 2.5 percent of scale length.

3.7.6 Instruction book. - Instruction books shall be furnished in accordance with Specification MIL-M-16616.

3.8 Voltage. - The peak ripple voltage at ohmmeter terminals shall be not more than 10 percent of the effective voltage. When the ohmmeter is tested as indicated on the instruction plate (see 4.6.1.1 and 4.6.1.2), the voltage at the connection terminals shall be not less than 400 nor more than 600 volts d.c. When the ohmmeter is tested as specified in 4.6.1.4, the voltage shall be not less than 400 volts nor more than 600 volts d.c.

3.9 Heat effect at 65°C. - When tested in accordance with 4.6.4 the ohmmeter shall indicate freely the heat effect at 65°C. and the change in indication shall not exceed 4 percent maximum at 65°C. with 2 percent maximum permanent change.

3.10 Temperature influence. - The temperature influence shall not exceed 1.5 percent maximum.

3.11 Low temperature exposure. - During exposure to a temperature of 0°C., ohmmeters shall indicate freely. The change in indication at the low temperature, and the change in indication shall not exceed 3 percent maximum change with 1 percent maximum permanent change.

3.12 External- field influence. - The ohmmeters shall have sufficient magnetic shielding so that the effect of an external magnetic field of 5 oersteds will not cause a change in indication greater than 1.5 percent.

3.13 Dielectric strength. - When tested in accordance with 4.6.8 ohmmeters shall withstand a voltage of 1000 volts d.c. without damage or flashover.

3.14 Humidity resistance. - After the test specified in 4.6.9 ohmmeters shall operate freely, and the permanent change in indication shall not exceed 2 percent of scale length.

3.15 Identification of product. - Each ohmmeter shall be marked for identification in accordance with Standard MIL-STD-130.

3.16 Workmanship. - The ohmmeters shall be fabricated in accordance with 3.1 thru 3.3.3 and shall be free from burrs, chips and surface defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Acceptance inspection.
    1. Acceptance inspection of equipment before preparation for delivery (see 4.4).
    2. Acceptance inspection of preparation for delivery (see 4.5).

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4.3 Preproduction inspection. - The preproduction samples shall be submitted to a laboratory satisfactory to the bureau or agency concerned. These samples shall be subjected to the examination and tests specified in 4.4 and 4.6 and any other tests considered necessary to determine conformance with this specification. When a manufacturer can present proof of a previous approval of identical ohmmeters, pre-production inspection may be waived by the procuring activity for the next 2500 ohmmeters or a period of 5 years wichever occurs first. After this period, the manufacturer will be required to submit a preproduction sample for test.

4.4 Acceptance inspection of equipment before preparation for delivery. - This inspection shall consist of groups A and B.

4.4.1 Inspection lot. - An inspection lot shall consist of ohmmeters of the same type, produced under the same conditions and offered for inspection at one time.

4.4.2 Group A inspection. - This inspection, including sampling, shall conform to Standard MIL-STD-105. Examination and tests and their acceptable quality level (AQL) shall be in accordance with Table I.

Table I - Group A Inspection

Examination or test	Reqt Par.	Method Para.	Acceptable Quality Level (AQL)	
			Major	Minor
Visual and mechanical	3.16	4.4.2.1	1.5 DPHU*	6.5 DPHU*
Electrical				
Initial accuracy	3.6.5 3.7.5	4.6.2	1.0%	....
Dielectric strength	3.13	4.6.9	*combined*	....

*Defects per hundred units

4.4.2.1 Classification of defects. - Defect as listed and classified in Table II shall be used as criteria for rejection during visual and mechanical inspection. Table II shall not be considered as listing all critical, major and minor defects.

4.4.3 Group B inspection. - This inspection, including sampling, shall conform to the special procedures for small sample inspection of Standard MIL-STD-105. Normal inspection shall be used at the start of the contract. Small sample reduced inspection shall be R-1. Examinations, tests and sampling plan shall be in accordance with Table III.

4.5 Acceptance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirement in Section 5.

4.6 Test procedures. -

4.6.1 Voltage. -

4.6.1.1 High resistance test. - The ohmmeter shall be operated as directed on the instruction plate with a resistance equivalent to the largest megohm reading for which the ohmmeter is designed in series with a low resistance multirange precision direct current microammeter, the combination being placed across the terminals of the meter. The product of the current as indicated by the microammeter and the value of the series resistance represents the voltage at the connection terminals of the ohmmeter.

4.6.1.2 Low resistance tests. - The ohmmeter shall be operated as directed on the instruction plate with a resistance of 1 megohm if type GC, GM, PC, and 10 megohms if type PM in series with a low resistance multirange precision direct current microammeter, the combination being placed across the terminals of the ohmmeter. The product of the current as indicated by the microammeter and the value of the series resistance represents the voltage at the connection terminals of the ohmmeter.

4.6.1.3 Ripple voltage. - The determination of ripple voltage shall be made at the ohmmeter terminals by means of oscillographic measurements of the currents for the arrangements specified in 4.6.1.1 and 4.6.1.2 (see 3.8).

4.6.1.4 Constant voltage. - Tests shall be made at various cranking speeds above nominal to determine compliance with 3.4.9 and 3.8.

4.6.2 Initial accuracy. - A calibration test shall be made to determine compliance with 3.6.5 and 3.7.5. Readings shall be taken at 10 equidistant points on the scale including a reading at the end scale value.

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Table II - Classification of defects in accordance with Standard MIL-STD-105.

Categories	Defects
Critical:	
1	Construction such that personnel may receive electric shock in operation.
Major:	
101	Type not as specified.
102	Material nonconforming; evidence of unauthorized materials used.
103	Incomplete, component parts missing.
104	Component parts not new; evidence of used, reclaimed, repaired or salvaged parts.
105	Corrosion resisting material not used for metal parts, fittings, and screws.
106	Magnets, steel and iron parts in magnetic circuit not treated to resist corrosion.
107	Designated coils not treated with three impregnating coats of varnish as specified.
108	Circuit elements designated not coated with moisture and fungus resistant material as specified.
109	Identification plates nonconforming; not engraved; not readily visible.
110	Case not specified material; not dust-tight; all parts, except crank, not enclosed.
111	Indicating meter mechanism not enclosed in dust-tight compartment forming part of case.
112	For type GC or GM specified type handle not provided; crank, if provided, protrudes farther from case than specified.
113	Other end test leads not provided with brass spade clip.
114	Indicating meter window cloudy; scale or pointer obscured.
115	Threaded parts roughly finished; loose or poor connections; lock washers not provided.
116	Specified number of binding posts and test leads not provided.
117	Generators not smooth running; excessive starting and running torques.
118	Instruction plate not specified type, material; not securely mounted in case.
119	Instruction plate does not contain all specified information.
Minor:	
1	Test leads of specified length not furnished; leads do not terminate in insulated test clip.
2	Spade clip not provided with insulating sleeve over cord connection joint.
3	One direction armature, if used, not provided with specified ratchet or other device, to prevent wrong way rotation.

Table III - Group B inspection

Examination or test	Reqt. par.	Method Par.	Sampling plan
Low voltage resistance	3.5	4.6.1.2	AQL - 4.0%
High voltage resistance	3.5	4.6.1.1	L-8 normal & tightened
Constant voltage	3.5	4.6.1.4	L-6 reduced

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4.6.3 Effect of external magnetic fields on meter accuracy. - Suitable tests shall be made with alternating 60 cycle and unidirectional magnetic fields to determine compliance with 3.12.

4.6.4 Heat effect at 65°C. - Ohmmeters shall be maintained at 20°C ± 5°C for a sufficient time (not less than 2 hours) to attain thermal stability, and shall then be subjected to the three successive cycles of temperature change listed in table IV. During this test, ohmmeters shall indicate freely at all parts of the scale.

Table IV - Procedure for determining heat effect at 65°C.

Cycle number	Period (hours)	Step number	Procedure
-	--	1	Take initial reading at 20°C ± 5°C.
-	16	2	Place ohmmeter in chamber at 65°C. and maintain that temperature for the remainder of that period.
-	-	3	Read ohmmeter while at 65°C. at the end of step 2.
-	8	4	Allow chamber and ohmmeter to cool down to the 20°C ± 5°C during this period.
1	-	5	Read ohmmeter while at 20°C ± 5°C at the end of step 4.
-	16	6	Raise temperature of chamber and ohmmeter as rapidily as possible to 65°C. and maintain that temperature for the remainder of the period.
2	-	7	Read ohmmeter while at 65°C. at the end of step 6.
	8	8	Allow chamber and ohmmeter to cool down to the 20°C ± 5°C during this period.
-	-	9	Read ohmmeter while at 20°C ± 5°C at the end of step 8.
-	16	10	Raise temperature of chamber and ohmmeter as rapidily as possible to 65°C. and maintain that temperature for the remainder of the period.
-	--	11	Read ohmmeter while at 65°C. at the end of step 10.
3	8	12	Allow chamber and ohmmeter to cool down to the 20°C ± 5°C during this period.
-	-	13	Read ohmmeter while at 20°C ± 5°C at the end of step 12.

4.6.5 Temperature influence. - Ohmmeters shall be tested at temperatures 20°C. above and below the 20°C ± 5°C. Readings shall be taken at not less than four points on the scale, one of which is at or near the maximum scale value, and the temperature influence computed. The temperature influence shall be considered one-half the percentage change in indication caused by the 20°C. departures from the 20°C ± 5°C. If the influences above and below the 20°C ± 5°C are not equal, the greater value shall be considered the temperature influence.

4.6.6 Low temperature exposure. - Ohmmeters shall be exposed to a temperature of 0°C. for 16 hours, and at the end of the period, meter readings shall be taken. It shall be observed whether the ohmmeter indicated freely at all parts of the scale while 0°C. Ohmmeters shall then be maintained at 20°C ± 5°C for sufficient time to attain thermal stability, and meter readings repeated. The permanent change in indication resulting from low temperature exposure shall be noted. Readings shall be taken at not less than 5 points on the meter scale including a reading at the end scale value.

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4.6.7 Life (types GC and GM).- The ohmmeter shall be operated at a rated speed for 24 hours, actual running time, with a resistance of 1 megohm connected across the terminals. The test shall be made by operating the ohmmeter 3 hours per day in 5-minute periods with 5-minute intervals.

4.6.8 Dielectric strength.- A potential of 1,000 volts d.c. shall be applied between the current-carrying parts and the frame of the meter for 60 seconds. The leakage current as measured at the end of this period shall not exceed 0.001 milliampere. As an alternate to actual measurement of current, an insulation reading of not less than 1000 megohms will be acceptable.

4.6.9 Humidity resistance.- The ohmmeter shall be operated at 20°C ± 5°C and a calibration check made. The ohmmeter shall then be placed in a humidity chamber and kept for 6 hours at a temperature of 65°C. with a relative humidity of 90 percent. At the end of the 6 hour period, the heat shall be shut off. The ohmmeter shall then be permitted to cool for 18 hours in this atmosphere, the relative humidity of which rises as the temperature is reduced. A temperature of 35°C., or lower, but not less than 25°C. shall be attained during the cooling period. The complete cycle shall then be repeated. Upon completion of the second cycle the ohmmeter shall be removed from the humidity chamber and permitted to dry for a period of 24 hours at 20°C ± 5°C, with a relative humidity less than 60 percent. The ohmmeter shall then be operated and a calibration check made to verify compliance with 3.14.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- The equipment, accessories, technical publications and repair parts shall be preserved and packaged by level A or C; packed by level A, B or C and marked as specified (see 6.2) in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended use.-

6.1.1 Types GC and GM ohmmeters are intended for use in determining the insulation resistance of electric circuits and equipment where reasonable accuracy and fair portability are required. Type GC is primarily for use with shipboard power circuits and type GM is primarily for use with shore circuits.

6.1.2 Types PC, PM ohmmeters are intended for use in determining the insulation resistance of electric circuits and equipment where reasonable portability is required. Type PC is primarily for use with shipboard power circuits and type PM is primarily for use with ship and shore electronic circuits.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Preproduction samples required (3.1).
(d) Applicable moisture and fungus resistant treatment (see 3.2.3.1).
(e) Selection of the level of packaging and the level of packing required (see 5.1).

6.3 Superseding data.- This specification supersedes Specification MIL-O-16485B of 14 May 1956 (In part). This specification covers types GC, GM, PC and PM ohmmeters of Specification MIL-O-16485B(SHIPS).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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